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FLOW HIVE – A NOVEL TECHNOLOGY FOR BEEKEEPING

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The Flow Hive is a groundbreaking innovation in beekeeping, designed to make beekeeping simpler, safer and more efficient. A flow hive is a relatively new style of hive that uses standard Langstroth frames. This technology was developed by Australian inventors, Stuart and Cedar Anderson in 2015. It revolutionizes honey extraction while ensuring the well-being of the bees. Unlike traditional methods, which require removing frames, uncapping the honey, and using centrifugal force to extract it, the Flow Hive allows beekeepers to harvest honey directly from the hive without disturbing the bees or damaging the hive. The system features preformed, partially constructed honeycomb frames. Instead of removing the entire frame to harvest the honey, a tube is inserted into one end of the super. Once a flow key is inserted and turned 90°, the cells split causing honey to flow into the beekeeper's vessel, eliminating the need for handling the bees or disrupting the colony, making the entire process easier and less stressful for the bees. This hive is suitable for domestication of Italian bee (*Apis mellifera*) that are domesticated in Langstroth hive.

Components of a Flow Hive

Flow Hives are made from sustainable A Grade wood. It comprises various parts that resemble the conventional bee hive (Fig: 1).

1) Super Box

The super is the top part of the hive where the Flow frames are placed. It is usually a wooden box made of timber that holds the Flow frames in place.

2) Flow frames

Flow frames are typically made from food-grade plastic, which is durable and easy to clean. They specially designed, pre-formed honeycomb structures that allow honey to be extracted directly from the hive without disturbing the bees. Each frame consists of partially-

formed, segmented cells that can be opened by turning a lever. This creates a pathway for honey to flow out and into collection jars.

3) Frame holders

Frame holders are the wooden or plastic supports that hold the Flow frames in place within the super. These holders ensure the frames stay securely aligned and make it easy to remove or adjust the frames when needed.

4) Flow Hive Taps

These are the control mechanisms attached to the Flow frames. They allow beekeepers to open the frames by turning a knob or lever, releasing the honey to flow out of the combs. Each frame has its own tap. By turning the taps, the bees' honeycomb cells break open, and honey begins to flow into the hive's extraction system.

5) Hive Base/Bottom Board

The bottom board is the base of the hive that sits on the ground or on a stand. It has a small entrance for bees to come and go, and it provides structural support for the rest of the hive components. The bottom board can include a screen for ventilation or a solid surface. It helps ensure proper airflow and drainage within the hive.

6) Brood Box

The brood box is where the queen bee lays eggs and where the brood (young bees) are raised. It's located below the super box in a typical hive setup. The brood box contains frames for the bees to build brood combs and store pollen.

7) Frame Holders

Frame holders are the wooden or plastic supports that hold the Flow frames in place within the super. These holders ensure the frames stay securely aligned and make it easy to remove or adjust the frames when needed.

8) Ventilation/Entrance System

This system is used for ventilation or temperature regulation, such as entrance reducers or ventilation holes. These systems help maintain the health of the hive and ensure the bees are not stressed by extreme temperatures or excessive humidity.



Fig. 1: Image of authentic flow hive

9) Roof/Cover

The roof or cover is placed on top of the super box to protect the hive from the elements, such as rain or direct sunlight. It also helps to maintain the internal temperature of the hive.

Advantages of Flow Hive

The innovative design of a Flow Hive offers some significant benefits for the beekeeper and the honey bees.

1. **Easy honey harvesting:** Quick and easy honey extraction from the frames is the biggest benefit a Flow Hive.
2. **Minimal disruption:** Bees remain calm and undisturbed during harvesting.
3. **Increased safety:** Reduced risk of bee stings and attacks.
4. **Bee-friendly extraction:** Rather than disturbing the bees and lifting out a whole super, the hive remains intact during honey production. This non-invasive approach doesn't require brushing away bees or smoking.
5. **Minimized tools usage:** Tools like honey extractors can cost hundreds or even thousands, depending on the size of your operation. With flow hives, there is no need for these additional expenses.
6. **Hive efficiency:** Traditional honey harvesting may require the beekeeper to cut off the cappings and remove the honey, then replace the comb in the hive. This process causes massive cell damage which takes the hive a long time to repair.

Conclusion

Flow Hive promotes better hive management by minimizing the stress on bee colonies during harvests, which is crucial for maintaining healthy and sustainable bee populations. Whether for hobbyists or professionals, the Flow Hive has proven to be an accessible and effective tool for beekeepers around the world, helping to make honey harvesting cleaner, more environmentally friendly, and less disruptive to the delicate balance of the hive. Through its combination of innovation, simplicity and sustainability, the flow Hive is changing the way the world approaches beekeeping.

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